National University of Technology
Computer Engineering Department
Circuit Analysis (CEN1010)
Spring 2025

Complex Engineering Problem (CEP)

Project Title: Fully Regulated Variable DC Power Supply

1. Objectives

The objective of this term project is to build a fully regulated variable DC power supply. This power supply uses transformer to step down the ac supply, diodes for rectification, capacitor as filter, regulator integrated circuit to regulate output voltage at desired value and a load resistor or device.

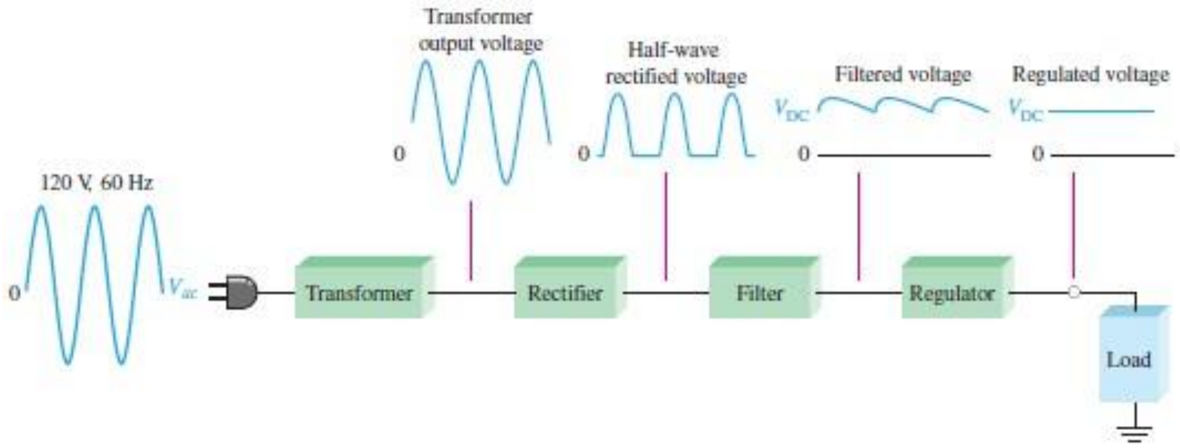

Figure 1. Schematic of complete power supply

2. Project Outcomes

The main features of the power supply are as follows:

1. It can supply the output voltage 1.2V to 30V at the max current of 1.5A.
2. Output Voltage Tolerance 1%
3. Line Regulation 0.01%
4. Load Regulation 0.3%
5. Prevent the deposition temperature
6. Short-circuit protection
7. Design a PCB for your prototype

Note: The due date of report submission is **Thursday May 22, 2025**. The project exhibition and presentation will be in **Monday May 26, 2025**

3. Project Description

3.1 Introduction

Almost all basic household electric and electronic circuits need an unregulated AC to be converted to constant DC, to operate the specific electronic device. All devices have a certain power supply limit and the electronic circuits inside these devices must be able to supply a constant DC voltage within this limit. This DC supply is regulated and limited in terms of voltage and current. However, the supply provided from mains power distribution line may fluctuate and could easily break down the sophisticated electronic equipment, if not properly limited. This work of converting an unregulated alternating current (AC) or voltage to a limited direct current (DC) or voltage to make the output constant regardless of the fluctuations in input, is done by a regulated power supply circuit.

Figure 2. A standard variable DC power supply

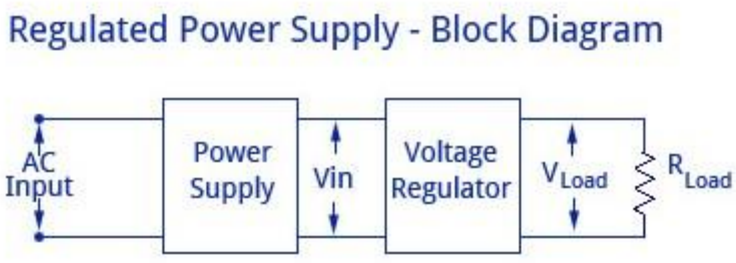

Figure 3. Block diagram of a regulated power supply

3.2 Procedure

There are various factors in designing the power supply like the load voltage, load current, voltage regulation, source regulation, output impedance, ripple rejection, and so on. You have to do the complete calculations as per the given outcomes then decide your hardware components and their values.

Note: Before implementing it on hardware, first you create a schematic on *Proteus software* based on your calculations and verity your outcomes. Then you implement it on hardware by designing a printed circuit board (PCB) for your prototype.

3.3 Key Factors

- **Load Regulation**: The load regulation or load effect is the change in regulated output voltage when the load current changes from minimum to maximum value.

$$\textbf{\% Load Regulation} = [(V_{no\text{-}load} - V_{full\text{-}load})/V_{full\text{-}load}] * 100$$

- **Minimum Load Resistance**: The load resistance at which a power supply delivers its full load rated current at rated voltage is referred to as minimum load resistance.
- **Source/Line Regulation:** Ratio of change in O/P voltage and change in I/P voltage.
- **Load Regulation:** Ration of change in O/P voltage to change in O/P current.
- **Ripple Rejection**: Voltage regulators stabilize the output voltage against variations in input voltage. Ripple is equivalent to a periodic variation in the input voltage. Thus, a voltage regulator attenuates the ripple that comes in with the unregulated input voltage. Since a voltage regulator uses negative feedback, the distortion is reduced by the same factor as the gain.

3.4 Order of Design

1) Load Calculation
2) Regulator
3) Bridge Rectifier
4) Transformer 5) Capacitor (Filter)

3.5 Tasks

1. Simulate the project on *Proteus* and extract the output waveforms at each stage.
2. Design the rectifier on hardware/PCB
   i) Do calculations for load voltage, load current, Line regulation, load regulation and capacitor value for filtering
   ii) Decide a voltage regulator IC as per the given outcomes.

3. **3.** Write a complete report on the project showing the output waveforms at each stage. (Report template is provided)

Note: You should use 2 A transformer with a full current up to the 1.5 A at the output. However, 1A transformer also works well with low current.

4. Policy on Professional Ethics & Plagiarism:

You are free to consult any book and online resources during the design and analysis phase, but you could not copy from them. Your design and implementation must be your original effort and same apply for simulations. Remember that if anyone is found coping from the Internet or other group members, the group shall face penalty. You are not allowed to copy any material or code directly from the web or elsewhere.

References

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